

1 Solution — GIAM 7.1

Problem 1

1.1 Problem

Determine the number of entries in the following sequences.

- a) (999, 1000, 1001, ..., 2006)
- b) (13, 15, 17, ..., 199)
- c) (13, 19, 25, ..., 601)
- d) (5, 10, 17, 26, 37, ..., 122)
- e) (27, 64, 125, 216, ..., 8000)
- f) (7, 11, 19, 35, 67, ..., 131075)

1.2 General Strategy

For each sequence we identify a closed form a_n in terms of an integer index n , determine the smallest and largest indices m, M that appear, then count

$$\# \text{entries} = M - m + 1.$$

For an arithmetic sequence $(a, a + d, a + 2d, \dots, L)$ with common difference $d > 0$, this specializes to the *fence-post formula*

$$\# \text{entries} = \frac{L - a}{d} + 1.$$

The "+1" matters: $(L - a)/d$ counts the *gaps* between consecutive entries, while the "+1" includes the first post that the gaps alone miss.

1.3 a) (999, 1000, 1001, ..., 2006)

Arithmetic with $a = 999$, $d = 1$, $L = 2006$:

$$\frac{2006 - 999}{1} + 1 = 1007 + 1 = \boxed{1008}.$$

1.4 b) (13, 15, 17, ..., 199)

Arithmetic with $a = 13$, $d = 2$, $L = 199$:

$$\frac{199 - 13}{2} + 1 = \frac{186}{2} + 1 = 93 + 1 = \boxed{94}.$$

1.5 c) (13, 19, 25, ..., 601)

Arithmetic with $a = 13$, $d = 6$, $L = 601$:

$$\frac{601 - 13}{6} + 1 = \frac{588}{6} + 1 = 98 + 1 = \boxed{99}.$$

1.6 d) (5, 10, 17, 26, 37, ..., 122)

The consecutive differences 5, 7, 9, 11, ... are themselves arithmetic, so the original sequence is quadratic. The first few entries reveal the pattern:

$$5 = 2^2 + 1, \quad 10 = 3^2 + 1, \quad 17 = 4^2 + 1, \quad 26 = 5^2 + 1, \quad 37 = 6^2 + 1.$$

Hence

$$a_n = n^2 + 1, \quad n = 2, 3, 4, \dots$$

The terminal entry satisfies $n^2 + 1 = 122$, i.e. $n^2 = 121$, giving $n = 11$. So n runs over $\{2, 3, \dots, 11\}$, and

$$11 - 2 + 1 = \boxed{10}$$

entries.

1.7 e) (27, 64, 125, 216, ..., 8000)

These are consecutive cubes: $27 = 3^3$, $64 = 4^3$, $125 = 5^3$, $216 = 6^3$, so

$$a_n = n^3, \quad n = 3, 4, 5, \dots$$

The terminal entry satisfies $n^3 = 8000 = 20^3$, so $n = 20$. Hence n runs over $\{3, 4, \dots, 20\}$, and

$$20 - 3 + 1 = \boxed{18}$$

entries.

1.8 f) (7, 11, 19, 35, 67, ..., 131075)

The consecutive differences are

$$11 - 7 = 4 = 2^2, \quad 19 - 11 = 8 = 2^3, \quad 35 - 19 = 16 = 2^4, \quad 67 - 35 = 32 = 2^5, \quad \dots$$

— successive powers of two. Indexing from $n = 1$ with $a_1 = 7$, the recurrence $a_n - a_{n-1} = 2^n$ telescopes:

$$a_n = a_1 + \sum_{k=2}^n 2^k = 7 + (2^{n+1} - 4) = 3 + 2^{n+1},$$

using the geometric sum $\sum_{k=0}^n 2^k = 2^{n+1} - 1$, so $\sum_{k=2}^n 2^k = 2^{n+1} - 4$.

Spot check: $a_1 = 3 + 4 = 7$, $a_2 = 3 + 8 = 11$, $a_3 = 3 + 16 = 19$, $a_4 = 3 + 32 = 35$, $a_5 = 3 + 64 = 67$. All match.

The terminal entry satisfies

$$3 + 2^{n+1} = 131075 \implies 2^{n+1} = 131072 = 2^{17} \implies n = 16.$$

Hence n runs over $\{1, 2, \dots, 16\}$, giving $\boxed{16}$ entries.

1.9 Summary

Part	Type	Closed form a_n	Index range	Count
a)	arithmetic, $d = 1$	$998 + n$	$1 \leq n \leq 1008$	1008
b)	arithmetic, $d = 2$	$11 + 2n$	$1 \leq n \leq 94$	94
c)	arithmetic, $d = 6$	$7 + 6n$	$1 \leq n \leq 99$	99
d)	quadratic	$n^2 + 1$	$2 \leq n \leq 11$	10

e)	cubic	n^3	$3 \leq n \leq 20$	18
f)	exponential, offset	$3 + 2^{n+1}$	$1 \leq n \leq 16$	16

Table 1.1